2026 EDITION

Network Security Laboratory

Session 01 · Lab Setup & Network Virtualization

Before We Start: Downloads

Download ALL files before starting the installation. Large files — start now if you haven't!

Cisco C7200 image (router)	~45 MB	Mega.nz (link on course page)
Cisco C3745 image (switch)	~38 MB	Mega.nz (link on course page)
Ubuntu 25.10 Cloud Image	~784 MB	cloud-images.ubuntu.com
Ubuntu Cloud Init Data ISO	~128 KB	github.com/GNS3/gns3-registry
Raspbian Desktop i386 (Darth)	~3.4 GB	downloads.raspberrypi.com

Credentials:

- **Alice / Bob** ubuntu / ubuntu
- **Darth** pi / raspberry
- **Cisco** no login

About the Instructor

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Course Overview — 2026 Edition

SESSION 01
GNS3 Lab Setup
Beginner

SESSION 02
Symmetric Crypto
Beginner

SESSION 03
TLS/SSL & Certificates
Intermediate

SESSION 04
Layer 2 Attacks
Intermediate

SESSION 05
WiFi Security & Cracking
Intermediate

SESSION 06
Layer 3 Attacks
Intermediate

SESSION 07
VPN & Phishing
Adv.

SESSION 08
SSL Strip & Passwords
Advanced

What We Will Do Today

1. Install and configure GNS3
2. Build a virtual network with router, switch, and 3 hosts
3. Achieve full connectivity (LAN + internet access)
4. Leave with a working lab environment for the rest of the course

This lab is the foundation for all future sessions.

What is GNS3?

NET

Graphical Network Simulator

Emulates real network devices — not a toy. Used by professionals to test configurations before deploying on real hardware.

IOS

Runs real Cisco IOS images

Uses Dynamips to emulate Cisco C7200 and C3745 hardware. The same IOS you would find in a physical rack.

VM

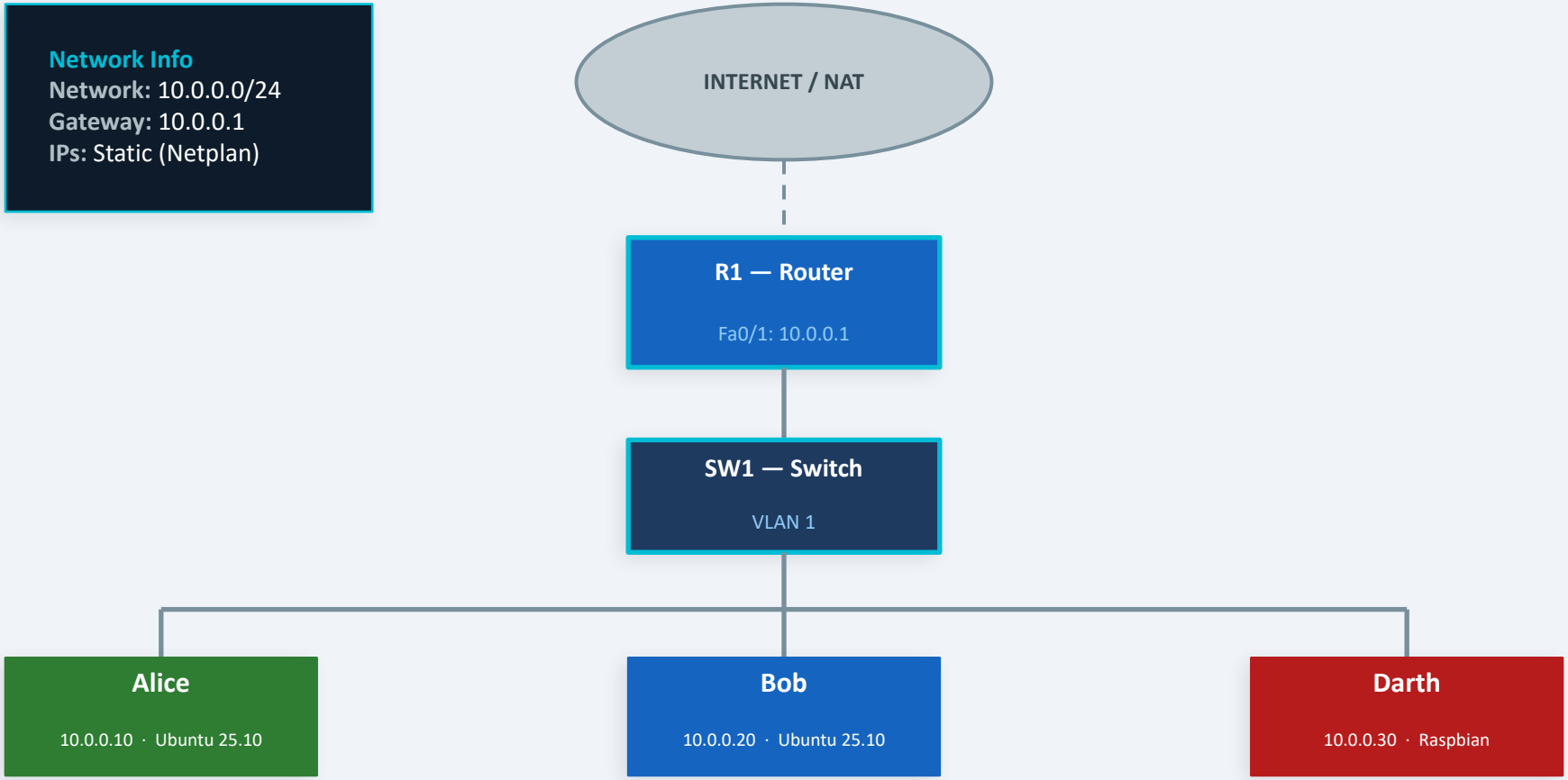
Ubuntu VMs as QEMU guests

Host machines (Alice, Bob, Darth) run as full virtual machines inside the simulation, connected through virtual switches.

Think of it as a virtual rack of real hardware.

Lab Topology

Network Info
Network: 10.0.0.0/24
Gateway: 10.0.0.1
IPs: Static (Netplan)



Why This Topology?

Alice & Bob

LEGITIMATE USERS

Will be targets in future labs. Their traffic, credentials, and communications will be intercepted, spoofed, and hijacked.

Darth

THE ATTACKER

Will launch Layer 2 and Layer 3 attacks from Session 04 onward — ARP spoofing, MAC flooding, IP spoofing, DNS hijacking.

R1 — Router

GATEWAY + NAT

Standard home/office router behaviour. Models real-world network edge. Provides NAT for internet access; hosts use static IPs.

SW1 — Switch

LAYER 2 HUB

Vulnerable to ARP spoofing and MAC flooding (Session 04). Connects all hosts to the same collision domain.

Every session builds on this same environment.

Components Overview

Component	Role	Technology
Cisco C7200	Router + NAT	**Dynamips emulator**
Cisco C3745	Layer 2 Switch	**Dynamips emulator**
Ubuntu 25.10	Alice & Bob (victim hosts)	**QEMU VM**
Raspbian (x86)	Darth (attacker host)	**QEMU VM**
Cloud node	Internet access (NAT out)	**virbr0 / bridge9**

Installation — Ubuntu Path

UBUNTU

1 Install GNS3 and dependencies

Add the GNS3 PPA repository to apt, then install: **gns3-gui, gns3-server, qemu-system, ubridge, kitty, tigervnc-viewer**
Add your user to groups: **ubridge, libvirt, kvm, wireshark**
When prompted → allow capturing without sudo and non-superuser access → both **Yes**
Log out and back in to apply group changes, then verify: gns3server --version

2 Configure NAT (internet access for VMs)

Enable **libvirtd.socket** — creates the virtual bridge **virbr0 (192.168.122.1)**
Add iptables rules: NAT masquerade on your internet interface, forward RELATED/ESTABLISHED traffic, forward virbr0 traffic
Save rules with iptables-persistent · Enable IP forwarding persistently via /etc/sysctl.conf

3 Verify install and launch GNS3

Verify installation: run **gns3server --version**
Launch GNS3 → Setup Wizard → **"Run on local computer" → localhost:3080**

macOS users: see next slide for a different installation path.

Installation — macOS Path

Apple Silicon

No GNS3 VM needed — gns3-server runs natively. ubridge and dynamips are compiled from source.

1 Install dependencies + GNS3 GUI (once)

Via Homebrew: qemu, cmake, libpcap, libelf, pkg-config, pipx
GNS3 GUI via brew cask · gns3-server via pipx

2 Build ubridge from source (once)

Clone github.com/GNS3/ubridge, compile with make, install system-wide
Verify: which ubridge

3 Build dynamips from source (once)

Clone github.com/GNS3/dynamips, create build directory
Configure with CMake (stable mode), compile and install system-wide

4 Fix TCP_KEEPALIVE bug (once — reapply after upgrades)

Find telnet_server.py inside the gns3-server pipx virtual environment
Replace TCP_KEEPIDLE with TCP_KEEPALIVE in that file

5 Start server + launch GUI (every session)

Run gns3server in a terminal and keep it open
When ready → open GNS3.app → Wizard: Local computer → ~/.local/bin/gns3server

6 NAT setup — virtual bridge (every session, not persistent)

Create bridge9 + feth virtual interfaces, assign 192.168.122.1/24
Enable IP forwarding, configure pfctl NAT for your internet interface
macOS does not persist virtual interfaces — redo before each session

Device Templates

Create once — reused across all projects. All templates: File → New Template → Install from GNS3 server.

C3745

Cisco C3745 — Switch (Dynamips)

1. File → New Template → Install from GNS3 server → select c3745 image
2. Slot 1: NM-16ESW · startup config: C3745_startup-config.cfg · Symbol: switch

⚠ **Use FastEthernet1/0–1/15 for host connections. Fa0/0 / Fa0/1 are routing ports — avoid!**

C7200

Cisco C7200 — Router (Dynamips)

1. File → New Template → Install from GNS3 server → select c7200 image
2. Slot 0: C7200-IO-2FE · startup config: c7200_startup-config.cfg
3. Fa0/0 → NAT outside (192.168.122.x) · Fa1/0 → LAN (10.0.0.1) · Hosts use static IPs

Ubuntu

Ubuntu 25.10 — Alice & Bob (QEMU, GNS3 registry)

1. Resize the downloaded .img file (+10 GB) using qemu-img before importing into GNS3
2. File → New Template → Install from GNS3 server → search "ubuntu cloud" → Ubuntu Cloud Guest
3. Import: ubuntu-25.10-server-cloudimg-amd64.img + ubuntu-cloud-init-data.iso → Finish

Darth

Raspbian i386 — Darth (QEMU, GNS3 registry)

1. File → New Template → Install from GNS3 server → search "ras" → Raspian → Install
2. Right-click template → Configure → HDD tab → Create... → Qcow2 → 30,000 MiB

⚠ **First boot = OS installer (~15 min). Select 'Install' (not Graphical). Then reboot.**

Network Configuration — Hosts

Ubuntu (Alice & Bob): Netplan YAML · Raspbian (Darth): /etc/network/interfaces

Alice (10.0.0.10) · Bob (10.0.0.20) · ubuntu / ubuntu	
Config file	/etc/netplan/00-lab.yaml
Interface	ens3 · DHCP disabled
Alice IP	10.0.0.10 / 24
Bob IP	10.0.0.20 / 24
Gateway	10.0.0.1
DNS	8.8.8.8 and 8.8.4.4

chmod 600 on the file · delete 50-cloud-init.yaml · then netplan apply

Darth (10.0.0.30) · pi / raspberry	
Config file	/etc/network/interfaces
Interface	eth0 · static
IP address	10.0.0.30
Netmask	255.255.255.0
Gateway	10.0.0.1
DNS file	/etc/resolv.conf (locked with chattr +i)

Apply: systemctl restart networking · verify with ip addr show eth0

Cloud node → right-click → Configure → select virbr0 (Ubuntu) or bridge9 (macOS)

Verification

Start all nodes (right-click canvas → Start all nodes). Run these checks from each machine:

Alice & Bob · ubuntu / ubuntu	Darth · pi / raspberry
✓ **Ping the router** 10.0.0.1 — gateway reachable	✓ **Ping the router** 10.0.0.1 — gateway reachable
✓ **Ping Bob from Alice** 10.0.0.20 — LAN connectivity	✓ **Ping Alice from Darth** 10.0.0.10 — LAN connectivity
✓ **Ping Darth from Alice** 10.0.0.30 — LAN connectivity	✓ **Ping Bob from Darth** 10.0.0.20 — LAN connectivity
✓ **Ping 8.8.8.8** Internet access by IP	✓ **Ping 8.8.8.8** Internet access by IP
✓ **Check DNS resolver status** DNS: 8.8.8.8 and 8.8.4.4	✓ **Check /etc/resolv.conf** nameserver 8.8.8.8 and 8.8.4.4
✓ **Ping google.com** DNS resolution working	✓ **Ping google.com** DNS resolution working

All six checks must succeed on every machine.

Common Problems & Fixes

virbr0 missing in Cloud node

Ubuntu

Start the default virtual network with virsh, then set it to autostart on boot.

GNS3 500 error when adding images

Ubuntu

Disk is over 90% full — check available space and free up room, then retry.

Ubuntu VM disk full

Ubuntu

Close the VM first, then resize the qcow2 disk image (+8 GB), then restart.

Console connection fails (blank screen)

macOS

TCP_KEEPALIVE fix was not applied or server was running. Stop server, reapply the patch, restart.

bridge9 / feth0 disappear after reboot

macOS

Expected — macOS does not persist virtual interfaces. Re-run the NAT setup before each session.

GNS3 GUI can't connect to server

macOS

Start gns3server in a terminal first, wait for it to be ready, then open GNS3.app.

Homework

Verify your lab works before next session.

1. Start all devices in your GNS3 project
2. From Alice, ping the router, Bob, Darth, and the internet (8.8.8.8)
3. From Bob, ping Alice and the internet
4. Take a screenshot of successful pings

All checks must be done · If something fails, note the error and bring it to next session.

If something does not work, bring the error to next session.

What's Next

Session 02 · Symmetric Cryptography & Steganography

ENC	OpenSSL — encrypt and decrypt files and messages
IMG	Steganography — hide data inside images
NET	Encrypted channels with netcat and cryptcat
PY	Python crypto with pycryptodome library